

PUNTA GORDA, FL, USA

WMO: 722034

Lat: 26.917N

Lon: 81.991W

Elev: 25

StdP: 14.68

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 12812

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	38.7	42.9	26.7	20.7	51.3	30.4	24.5	52.1	24.1	69.4	20.9	68.3	5.7	20	0.379

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.7	93.4	76.5	92.4	76.5	91.2	76.5	80.1	87.4	79.3	87.0	78.7	86.4	8.3	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.4	147.4	82.3	77.4	142.5	82.1	76.8	139.7	81.9	43.3	87.0	42.7	87.0	42.1	86.8	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.1	18.0	16.2	DB	31.8	96.0	3.0	1.7	29.7	97.2	27.9	98.2	26.3	99.2	24.1	100.5	(4)
(5)			WB	29.5	81.5	3.4	1.2	27.0	82.3	25.0	83.1	23.1	83.7	20.7	84.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	73.9	61.8	65.2	68.3	72.8	77.4	81.4	82.6	82.9	81.7	76.8	69.5	65.6		
	DBStd	8.85	7.72	7.04	5.86	4.46	3.30	2.23	1.91	1.89	2.18	4.91	5.95	7.46		
	HDD50	14	9	2	0	0	0	0	0	0	0	0	0	3		
	HDD65	392	152	78	38	5	0	0	0	0	0	4	27	88		
	CDD50	8729	376	427	567	684	850	942	1009	1021	950	830	585	488		
	CDD65	3634	54	84	140	240	385	492	544	556	500	369	162	108		
	CDH74	33958	388	662	1199	2258	3938	4796	5463	5666	4634	3110	1152	692		
	CDH80	13044	39	112	254	739	1690	2145	2329	2392	1870	1116	242	116		
(14) Wind	WSAvg	7.3	7.9	7.8	8.5	8.3	7.8	6.6	5.7	5.8	6.3	7.5	7.6	7.5		
(15) Precipitation	PrecAvg	51.5	2.2	2.0	2.5	2.3	3.0	9.3	8.0	8.3	7.6	3.2	1.5	1.5		
	PrecMax	74.4	10.8	4.1	8.7	5.7	7.7	18.9	13.0	14.0	14.4	10.9	5.2	5.6		
	PrecMin	35.3	0.0	0.0	0.2	0.3	0.1	4.6	4.6	4.6	4.5	0.6	0.0	0.2		
	PrecStd	8.1	2.2	1.2	1.9	1.5	2.0	3.7	2.5	2.3	2.6	2.4	1.3	1.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	83.4	85.9	86.9	90.7	93.5	95.0	94.4	94.9	93.2	91.7	87.4	85.4		
		MCWB	70.1	70.2	69.4	72.1	72.3	75.7	77.1	77.9	76.9	76.3	75.0	73.1		
	2%	DB	81.0	83.2	84.5	88.4	91.4	93.1	92.9	93.0	91.6	89.8	84.8	83.0		
		MCWB	68.7	70.0	69.2	71.3	72.5	75.9	77.3	77.9	77.0	75.8	73.0	71.7		
	5%	DB	78.8	81.2	82.8	86.6	90.1	91.3	91.4	91.6	90.5	88.1	82.7	81.0		
		MCWB	67.7	68.7	69.2	70.5	72.5	76.0	77.4	77.8	77.0	74.9	71.2	70.5		
	10%	DB	75.4	78.8	81.1	84.3	88.2	90.0	90.2	90.4	89.2	86.2	80.9	78.5		
		MCWB	65.9	67.6	68.3	69.7	72.5	76.0	77.5	77.9	76.9	74.0	69.5	69.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.1	73.8	75.8	76.5	77.4	80.2	80.4	81.1	80.5	78.9	77.4	75.8		
		MCDB	79.0	80.8	81.9	84.9	87.1	88.1	88.0	87.7	87.1	85.5	83.4	81.8		
	2%	WB	71.1	72.4	73.9	75.0	76.4	78.8	79.4	80.2	79.4	78.0	75.4	74.0		
		MCDB	77.1	79.5	79.7	83.7	85.7	87.4	87.7	87.3	86.6	85.1	81.9	79.5		
	5%	WB	69.6	70.9	72.4	73.5	75.5	78.1	78.8	79.4	78.6	77.2	73.7	72.4		
		MCDB	75.8	77.8	78.5	81.6	84.4	86.8	87.0	86.9	85.9	84.3	79.5	77.9		
	10%	WB	68.0	69.4	70.6	72.3	74.8	77.4	78.3	78.8	78.1	76.3	72.0	70.6		
		MCDB	73.7	76.1	77.2	80.4	83.5	86.0	86.3	86.3	85.4	83.0	77.8	76.1		
(35) Mean Daily Temperature Range	MDBR		22.7	22.9	23.0	23.1	22.3	17.7	15.9	15.7	15.6	18.5	21.1	21.2		
	5% DB	MCDBR	23.2	23.2	23.9	24.6	24.4	20.4	17.5	17.2	17.1	18.5	20.5	21.0		
		MCWBR	12.1	11.4	10.7	10.1	8.7	6.9	5.7	5.5	5.5	6.6	9.3	10.5		
	5% WB	MCDBR	20.0	20.0	18.9	20.7	20.0	16.7	15.7	15.4	15.2	15.8	17.6	18.0		
		MCWBR	12.0	10.9	10.5	10.2	8.0	6.5	5.7	5.7	5.5	6.2	9.1	10.3		
(40) Clear-Sky Solar Irradiance	taub		0.347	0.361	0.375	0.391	0.419	0.443	0.461	0.448	0.422	0.397	0.359	0.350		
	taud		2.512	2.481	2.431	2.359	2.336	2.347	2.291	2.355	2.404	2.443	2.493	2.528		
	Ebn at Noon		286	290	292	288	278	269	264	268	273	274	279	280		
	Edh at Noon		30	33	36	40	41	40	42	40	37	33	30	28		
(44) All-Sky Solar Radiation	RadAvg		1166	1420	1745	2043	2119	1908	1862	1796	1623	1506	1259	1077		
(45)	RadStd		85	85	119	87	135	166	121	98	99	89	65	78		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+1.13	+2.61	+3.35	N/A	N/A	N/A	-7	N/A	+446	+382
Regional (31 neighbors)		+0.76	+1.48	N/A	N/A	N/A	N/A	N/A	-63	+284	+209

Nomenclature: See separate page